

User Guide: Bubble Sort Visualizer

Daksh Agarwal

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1 Introduction

Welcome to the Bubble Sort Visualizer! This is a simple command-line program that sorts a list of numbers you provide in a file. Its main feature is showing you the state of the list after every single "pass" of the sorting algorithm, which helps in understanding how Bubble Sort works.

This guide will walk you through setting up the program, running it, and understanding its output.

2 System Requirements

To use this program, you will need the following software installed on your computer:

- **A C Compiler:** The program must be compiled. The most common compiler is **GCC**.
- **A Plain Text Editor:** To create the source code and input files (e.g., Notepad, VS Code, TextEdit).

3 Setup and Execution

Follow these four steps to get the program running.

Step 1: Create the Program File

Open your text editor, copy the entire C source code for the program, and save it in a new folder with the exact name `bubble_sort.c`.

Step 2: Create the Input File

In the **same folder**, create another new file named exactly `input.txt`. Open this file and type the numbers you want to sort, separated by spaces.

Example `input.txt` content:

```
8 5 2 9 5 6 3
```

Your project folder should now contain two files:

```
MyProject/  
  bubble_sort.c  
  input.txt
```

Step 3: Compile the Program

Open your terminal (Command Prompt on Windows, Terminal on macOS/Linux) and navigate to the folder containing your files. Execute the following command:

```
gcc bubble_sort.c -o bubble_sort_app
```

This command compiles your code. If no errors appear, an executable file will be created in your folder. The process will look like this:

```
$ ls  
bubble_sort.c  input.txt  
  
$ gcc bubble_sort.c -o bubble_sort_app  
  
$ ls  
bubble_sort_app  bubble_sort.c  input.txt
```

Listing 1: Simulated terminal session for compilation.

Step 4: Run the Program

In the same terminal window, run the executable you just created:

4 Understanding the Output

When you run the program, it will print the results directly to your terminal. Below is the full example output for the input 8 5 2 9 5 6 3.

```
Attempting to read from input.txt...  
Successfully read 7 integers.  
  
Initial unsorted array:  
8 5 2 9 5 6 3  
  
Bubble Sort Passes Matrix:  
Pass 1: 5 2 8 5 6 3 9  
Pass 2: 2 5 5 6 3 8 9  
Pass 3: 2 5 5 3 6 8 9  
Pass 4: 2 5 3 5 6 8 9  
Pass 5: 2 3 5 5 6 8 9  
Pass 6: 2 3 5 5 6 8 9  
Pass 7: 2 3 5 5 6 8 9
```

Listing 2: Example program output.

The output is divided into two main parts:

- **Initial unsorted array:** This line shows the list of numbers exactly as they were read from your `input.txt` file.
- **Bubble Sort Passes Matrix:** This is a table that visualizes the sorting process.
 - Each row, labeled "Pass X", represents the state of the array after one full round of the Bubble Sort algorithm.
 - You can observe how the largest numbers "bubble up" to the end of the list with each pass.
 - In the example, the list becomes fully sorted by Pass 5. The subsequent passes show no change, confirming the array is sorted.

5 Troubleshooting

If you encounter an issue, refer to the common problems below.

Error: Error opening input.txt... This means the program cannot find the input file. Ensure `input.txt` is named correctly (all lowercase) and is in the same directory as the executable.

Error: gcc: command not found This means a C compiler is not installed or not available in your system's PATH. You must install a compiler like GCC (e.g., via MinGW for Windows, Xcode Command Line Tools for macOS, or 'build-essential' on Linux).

No output or "0 integers read" This indicates your `input.txt` file is empty or does not contain valid numbers. Please check the file's content.